# EVALUATING WEIGHT-TWO FORMS ON SHIMURA CURVES BY THE SHIMIZU LIFT, AND WHAT THE VALUES DECIDE

Abstract. We record how the differentials of an Atkin–Lehner quotient $X_0^D(N)/W$ are evaluated at points of the upper half-plane by Nelson's formula for the Shimizu lift, how the formula is extended from the newforms of level exactly $DN$ to the whole space of differentials, and the two uses made of the values: Noether's count of quadrics through the canonical image, which decides hyperellipticity numerically, and, in genus three with an Atkin–Lehner involution whose quotient has genus one, an explicit hyperelliptic model recognised from three rational invariants and confirmed by point counts. We state precisely what each step proves and what it does not, and we record the two traps met on the way: the working precision must rise with the cutoff of the lattice sum, and the Eichler order that Magma returns is not the same from one process to the next.

## 1. The formula

Let $B$ be the indefinite quaternion algebra over $\mathbb{Q}$ of discriminant $D$, let $R$ be an Eichler order of level $N$ with $(D,N)=1$ and $DN$ squarefree, and let $X = X_0^D(N)$ be the Shimura curve $\Gamma\backslash\mathbb{H}$ for the group $\Gamma$ of units of reduced norm one, embedded in $\mathrm{SL}_2(\mathbb{R})$ by a fixed splitting $\iota$. A weight-$k$ form $f_B$ on $X$ with Jacquet–Langlands correspondent a newform $f=\sum a_f(n)q^n$ of level exactly $DN$, $a_f(1)=1$, is given by the following theorem of Nelson [1, Theorem 3.1], which is the Shimizu lift made explicit. For $z\in\mathbb{H}$ put $\sigma_z = \left(\begin{smallmatrix} y^{1/2} & xy^{-1/2} \\ 0 & y^{-1/2}\end{smallmatrix}\right)$, so that $\sigma_z i = z$, and for a real $2\times 2$ matrix put

$$X\begin{pmatrix} a & b \\ c & d\end{pmatrix} = \frac{a+d+i(b-c)}{2}, \qquad P\begin{pmatrix} a & b \\ c & d\end{pmatrix} = \frac{a^2+b^2+c^2+d^2}{2}.$$

For $d \mid DN$ let $R^{(d)} = R + \frac{DN}{d}R^\vee$, where $R^\vee$ is the dual of $R$ for the trace form; it is a lattice between $R$ and $R^\vee$ of index $d^2$ over $R$, and the reduced norms of its elements have denominators dividing $d$. Then for $z_1, z_2 \in \mathbb{H}$,

$$(y_1y_2)^{k/2}\,\overline{f_B(z_1)}\,f_B(z_2) = \sum_{d\mid DN} \frac{\mu((d,N))}{d\,a_f(d)} \sum_{\substack{\alpha\in R^{(d)} \\ \operatorname{nrd}\alpha>0}} a_f\big(d\,\operatorname{nrd}\alpha\big)\, e^{ik\arg X(\alpha_{12})}\, W_k\big(4\pi|X(\alpha_{12})|\big), \quad (1)$$

where $\alpha_{12} = \sigma_{z_1}^{-1}\iota(\alpha)\sigma_{z_2}$ and

$$W_k(x) = 2^{1-k}\sum_{n\ge 1} n^k\big(nx\,K_{k-1}(nx) - K_k(nx)\big).$$

Two features of (1) govern everything below.

First, the left side is a product of a value and a conjugate value. An individual value $f_B(z)$ is defined only up to a constant of modulus one, and the right side determines the product. Fixing $z_1$ once and reading $\overline{f_B(z_1)}f_B(z_j)$ across $j$ gives every form with its own constant in front, which for the purposes below (a basis of differentials up to rescaling of its members) costs nothing.

Second, the sum is finite once truncated, and the truncation is an enumeration of a positive definite quaternary form. Since $2|X(\alpha)|^2 = P(\alpha) + \det\alpha$, the condition $4\pi|X| \le c$ is implied by $P(\alpha) \le 2(c/4\pi)^2$, and $P$ is positive definite on $M_2(\mathbb{R})$. The function $W_k$ decays like $e^{-x}$, so the terms dropped at cutoff $c$ are of size about $e^{-c}$ relative to the largest; in practice the cutoff buys about 0.4 decimal digits per unit, with a scatter of a digit or so between bases and between points, and no trend in $DN$. There is no fundamental domain, no reduction and no sampling anywhere in the method.

The ingredients are therefore: the algebra and an Eichler order, a real splitting, the lattices $R^{(d)}$, the Fourier coefficients $a_f(n)$ for $n \le DN(c/4\pi)^2$, and $W_k$. The only transcendental ingredient is $W_k$, evaluated once per lattice point; it is tabulated once per cutoff by a Chebyshev approximation of $W_k(x)e^x$ on panels in geometric progression, the geometric spacing being what keeps the singularity of $W_k$ at the origin equally far from every panel.

**The hypothesis of the theorem.** Theorem 3.1 is stated for a newform of level exactly $DN$. The space of differentials of $X_0^D(N)$ is the $D$-new part of $S_2(\Gamma_0(DN))$, which also contains, for every proper divisor $M$ of $N$, the oldclass of each newform of level $DM$. Whether the newforms of level exactly $DN$ span the $W$-fixed differentials of a given quotient is a dimension count that takes a second, and it fails at a good many quotients of interest. The next section removes the restriction.

## 2. The oldform extension

Fix $M \mid N$, let $T$ be the set of primes dividing $N/M$, and let $g$ be a newform of level exactly $DM$ with correspondent $g_B$ on $X_0^D(M)$. Let $R_N \subset R_M$ be Eichler orders of levels $N$ and $M$, $R_M = R_N + MR_1$ for a maximal order $R_1 \supset R_N$. For $p \in T$ let $w_p$ be an element of $R_N$ of reduced norm $p$ normalising $R_N$ (an Atkin–Lehner element), and for $S \subseteq T$ let $w_S = \prod_{p\in S} w_p$. The weight-two slash operator

$$(f \mid w)(z) = \mathrm{nrd}(w)\, j(w,z)^{-2} f(wz)$$

is the pullback of the differential $f(z)\,dz$ under $z \mapsto wz$, and the oldclass of $g$ at level $N$ is spanned by the $2^{\#T}$ pullbacks $g_B \mid w_S$: $S = \emptyset$ is the natural projection $X_0^D(N) \to X_0^D(M)$ and $S = \{p\}$ the second degeneracy map at $p$. Any two choices of $w_p$ differ by a unit of $R_N$ of reduced norm one, up to sign, so $g_B \mid w_S$ does not depend on the choice.

**Proposition 2.1.** *On the oldclass of $g$ the Atkin–Lehner group acts as follows.*

1. *For $q \mid DM$, $w_q$ acts on every $g_B \mid w_S$ by the quaternionic eigenvalue $\varepsilon_q(g)$ of $g_B$ on $X_0^D(M)$, which is $a_g(q)$ for $q \mid D$ and $-a_g(q)$ for $q \mid M$.*
2. *For $p \in T$, $w_p$ sends $g_B \mid w_S$ to $g_B \mid w_{S\triangle\{p\}}$ with no scalar.*

*Consequently the eigenvectors are $v_\chi = \sum_{S\subseteq T} \chi(S)\, g_B \mid w_S$ for the characters $\chi$ of $\{\pm1\}^T$, and $w_m$ acts on $v_\chi$ by $\prod_{q\mid m,\ q\mid DM} \varepsilon_q(g) \prod_{p\mid m,\ p\in T} \chi(p)$.*

*Proof.* For (1), an Atkin–Lehner element $w_q$ of $R_N$ at a prime $q \mid DM$ lies in $R_M$ and normalises it, the two orders agreeing locally at $q$, so it is also an Atkin–Lehner element at level $M$ and acts on $g_B$ by its eigenvalue there; and $w_q w_S$ and $w_S w_q$ generate the same two-sided ideal of $R_N$, so they differ by a unit of $R_N$ of reduced norm one, on which the slash operator is trivial. For (2), $w_p^2$ has reduced norm $p^2$ and lies in $R_N$, and $w_p^2/p$ is a unit of $R_N$ of reduced norm one (this is checked exactly in the code for every $w_p$ used). The weight-two slash is trivial on scalars, by the factor $\mathrm{nrd}(w)$, and on units of norm one, which act trivially on forms for $\Gamma$. The sign convention at $q \mid D$ and $q \mid M$ is the one the dimension counts of the next paragraph confirm. □

The $W$-fixed differentials of $X_0^D(N)/W$ are then the $W$-fixed newforms of level exactly $DN$ together with one $v_\chi$ for each pair $(g,\chi)$ whose eigenvalue product is 1 for every $m \in W$. This count must equal the genus, and the code refuses to continue when it does not; agreement at every quotient tried, including classes with two primes in $T$, is the check of the eigenvalue bookkeeping, and in particular of the sign at $q \mid M$.

**Evaluating a pullback.** Formula (1) at level $M$ gives $S_g(z_1,z_2) = y_1 y_2\, \overline{g_B(z_1)}\, g_B(z_2)$. Since $\mathrm{Im}(wz) = \mathrm{nrd}(w)\, y/|j(w,z)|^2$,

$$y_1 y_2\, \overline{g_B(z_1)}\, (g_B \mid w_S)(z_2) = \frac{\overline{j(w_S,z_2)}}{j(w_S,z_2)}\, S_g(z_1, w_S z_2). \tag{2}$$

So every member of the oldclass is the level-$M$ evaluator at the moved point, times a phase, with the *same* constant $\overline{g_B(z_1)}$ in front for every $S$. Nothing is left to normalise: the relative scale of the pullbacks is fixed by the slash operator, and that is what makes $v_\chi$ an eigenvector with eigenvalue exactly $\chi(p)$ rather than some $c$ and $1/c$. The sign and scale are also measured directly: for each generator $m$ of $W$ and a few points $z$, the ratio of the value of every column at $w_m z$, times $\bar{j}/j$, to its value at $z$ is compared with the predicted eigenvalue, the rejected characters of each oldclass being carried along so that their predicted eigenvalues $-1$ are measured too. On every control the ratios are $\pm 1$ to the working accuracy.

**The Atkin–Lehner elements.** $w_m$ is found in the two-sided ideal $J_m = \{x \in R_N : \mathrm{trd}(xy) \in m\mathbb{Z} \text{ for all } y \in R_N\}$, the radical of $R_N/mR_N$, of index $m^2$, as an element of reduced norm exactly $m$; such an element generates $J_m$ and hence normalises $R_N$, which is checked directly as well. It is produced as a short vector of $J_m$ for the positive definite majorant of the real splitting. Magma's normaliser routines are for definite orders, and its principal-generator routine can return a generator of negative norm, so neither is used.

## 3. Sample points and cost

The values are taken at points of $\mathbb{H}$ spread by the square root of the arithmetic volume of $\Gamma$. The spread matters for the count of the next section: rows from nearby points are nearly proportional, and a matrix of quadratic monomials of effective rank near the number of columns always has a quadric through it. The symptom of clustered points is a smooth ramp of singular values instead of a gap, and a verdict is refused unless the gap reaches a stated size.

The cost is the lattice enumeration. The number of lattice points summed is, to within a few percent at every base measured,

$$10\left(\frac{c}{4\pi}\right)^4 \frac{\sigma_2(DN)}{DN}, \qquad \frac{\sigma_2(DN)}{DN} = \prod_{p \mid DN}\left(p + \frac{1}{p}\right) = DN \prod_{p \mid DN}\left(1 + \frac{1}{p^2}\right),$$

so the cost of one evaluation is linear in $DN$ at a fixed cutoff, with a constant that is larger when $DN$ has small prime factors, and grows like the fourth power of the cutoff. Several eigenforms cost one enumeration, not several: everything but $a_f(n)$ is shared, so a basis of $g$ differentials at a pair of points is one enumeration. The Fourier coefficients are needed up to $DN(c/4\pi)^2$, and the newform decomposition at level $DN$ is the first ingredient to become expensive, well beyond the levels of interest here.

## 4. Noether's count

Let $X$ have genus $g \geq 3$ and let $u_1, \dots, u_g$ be a basis of its differentials. The canonical image lies on $(g-2)(g-3)/2$ independent quadrics if $X$ is not hyperelliptic (Noether), and on $(g-1)(g-2)/2$ if it is, the image being then a rational normal curve. At genus three this is 0 against 1, at genus four 1 against 3. With values of the basis at $m$ points the quadrics through the image are the kernel of the $m \times \binom{g+1}{2}$ matrix of quadratic monomials, and the count is read off its singular values: rows and columns are normalised, the matrix is realified, and the number of relations is the position of the largest ratio between consecutive singular values, accepted only when that ratio is at least $10^4$ and the smallest singular value is below $10^{-4}$; otherwise, when nothing is near zero, there is no relation, and when the spectrum is a ramp the count is declared inconclusive.

**What a numerical kernel dimension proves.** Nothing, strictly: a small singular value is a small number. What the reading relies on is an observed separation. On every curve of this project the smallest singular value is either at the truncation level of the sum, which falls as the cutoff rises, or at least $10^{-2}$, which does not; nothing has been seen between $10^{-2}$ and $10^{-12}$. A verdict is therefore taken at two or more cutoffs and accepted as hyperelliptic only when the residual falls with the cutoff while the rest of the spectrum stays put, and as non-hyperelliptic only when it does not fall.

There is one more condition, which the first trap below explains: a residual that is stable across cutoffs can still be a rounding floor rather than a genuine singular value. The test that separates the two is the working precision. A genuine singular value does not notice it (on the non-hyperelliptic control the smallest singular value agrees to twenty-nine digits between thirty and fifty digits of precision), while a numerical zero moves with it (by a factor of about a hundred at the first base where the floor was met). So a stalled residual is settled by raising the precision, never by assumption.

The verdict is numerical. The project treats it as a strong indication that directs the exact methods (a model, or an exact argument that excludes a hyperelliptic involution), not as a proof.

## 5. The model

Let $X = X_0^D(N)/W$ have genus three and suppose that an Atkin–Lehner element $u \notin W$ has $X/\langle W, u\rangle$ of genus one. The canonical image of a hyperelliptic curve carries no moduli, so the values of the differentials alone cannot give the equation; one derivative does. With $x = u_2/u_1$ a coordinate on the canonical conic, the function $y = (dx/dz)/u_1(z)$ is odd under the hyperelliptic involution and $y^2 = f(x)$ with $f$ of degree eight. The derivative is taken by Cauchy's formula on a circle of radius $\rho$ around each sample point, where the trapezoidal rule converges like $(\rho/R)^n$, $R$ the distance to the nearest pole of $x$, and the noise of the input is amplified only by $1/\rho$; the same circle gives the derivative from half its points, which is the convergence check.

**The coordinate.** The hyperelliptic involution is central in the automorphism group, so $u$ descends to an involution of the conic and the coordinate can be chosen with $u\colon x \mapsto -x$. Concretely, $u$ fixes one of the three differentials and negates the other two; the conic has no cross term between the fixed one and the other two (this is checked, and it would fail if the eigenvalue convention were wrong), so it reads $q_{pp}e_p^2 +$ (binary form in $e_a, e_b$), and factoring the binary form gives $w_2^2 = w_1 w_3$ with $w_2 = e_p$. Then $x = w_2/w_1$, the octic is even, $f(x) = g(x^2)$,

and the quotient $X/u$ is $Y^2 = g(t)$. The evenness is read off the fit, not assumed: the quartic $g$ in $t = x^2$ has five coefficients, is fitted on five sample points, and must predict $y^2$ at the others.

**The invariants.** The coordinate-free invariants of a binary octic have heights far beyond what the values carry. The quartic $g$ with its two marked points $t = 0, \infty$ has instead the three numbers

$$A = \frac{c_0 c_2}{c_1^2}, \qquad B = \frac{c_1 c_3}{c_2^2}, \qquad C = \frac{c_2 c_4}{c_3^2},$$

unchanged by $t \mapsto \lambda t$ and by rescaling $g$, which fifteen digits settle at the control $X_0^{35}(1)$ ($A = 1939/952200$, $B = 34500/76729$, $C = 1939/5000$). Each is recognised by the classical certainty criterion: a fraction $p/q$ within $\varepsilon$ of the computed real part is accepted when $q^2\varepsilon < \frac{1}{2}$, so that no other fraction of denominator at most $q$ lies as close, with $\varepsilon$ taken from the imaginary part of the computed number, which is pure noise. Swapping the two roots of the binary form is $x \mapsto 1/x$, which reverses $g$ and exchanges $A$ and $C$; both orderings are legitimate.

**The twist.** $A, B, C$ fix $g$ up to $t \mapsto \lambda t$ and rescaling, so the curve is

$$y^2 = \kappa\, g_0(\mu x^2), \qquad g_0(t) = 1 + t + At^2 + A^2Bt^3 + A^3B^2Ct^4,$$

with $\kappa, \mu \in \mathbb{Q}^\times$ mattering only modulo squares, and a complex-analytic computation cannot see either: its coordinate differs from a rational one by a complex scalar, which moves $\kappa$ and $\mu$ by complex squares. Both are solved rather than searched. The genus-one quotient is $Y^2 = \kappa g_0(t)$, $\mu$ having disappeared under $t \mapsto t/\mu$, so at each odd prime $p$ of good reduction the point count of $Y^2 = g_0(t)$ over $\mathbb{F}_p$ against the trace formula for $X/\langle W, u\rangle$ gives the Legendre symbol $(\kappa \mid p)$, one linear equation over $\mathbb{F}_2$ in the exponents of $\kappa$ over the primes of $2DN$ and the bad primes of $g_0$; the system is heavily overdetermined and must have a unique solution. Then $\mu$ is read off the curve itself the same way, $\kappa$ being known. The resulting curve is put in reduced minimal form and its point counts over $\mathbb{F}_p$ and $\mathbb{F}_{p^2}$ compared with the trace formula.

**When the coordinate is not rational.** The two fixed points of $u$ on the conic are the points $x = 0, \infty$; they are either two rational points or a conjugate pair over a quadratic field $K$. In the second case the coordinate with $u\colon x \mapsto -x$ exists only over $K$, the quartic $g$ has coefficients in $K$, and the nontrivial element of $\mathrm{Gal}(K/\mathbb{Q})$ acts as $x \mapsto 1/x$, so that $B$ is rational while $A$ and $C$ are conjugate. This is what the values show at one base: $B$ recognised, $A$ and $C$ complex conjugates to sixteen digits, with the conic and the predictions as clean as anywhere. The descent of $y^2 = g(x^2)$ from $K$ to $\mathbb{Q}$ is not implemented, and the package reports the pattern rather than a failure to recognise.

**Where the fit is conditioned, and where it is not.** The quartic is determined at its large-$t$ end by the sample points with large $|x|$ and at its small-$t$ end by the others. When the moduli $|x|$ at the sample points span less than a factor of about five, the fit is ill conditioned and the invariants come out to far fewer digits than the values carry, with the predictions at $10^{-6}$ while the conic is at $10^{-15}$; this happened at several bases. The remedy is to add sample points where $|x|$ is large, which a coarse evaluation at a small cutoff finds, since $x$ is read off the conic.

## 6. The checks, and their status

Point counts determine the isogeny class of the Jacobian, not the curve, so four further checks are made, each of which the isogeny class cannot supply.

(a) The four fixed points of $u$ on $X$ are the images of the fixed points of the elements of the coset $uW$, CM points whose discriminants and multiplicities are known; their total over $\#W$ must be 4, as Riemann–Hurwitz demands for a double cover of a genus-one curve by a genus-three curve. On the model they are the points over $x = 0$, defined over $\mathbb{Q}(\sqrt{\kappa})$, and over $x = \infty$, defined over $\mathbb{Q}(\sqrt{\text{leading coefficient}})$; the fields of definition of the CM points on $X$ are computed independently, by Shimura reciprocity, and must match pair by pair. A discriminant carrying four points of $X$ defined over one quadratic field contributes that field twice.
(b) The Jacobian of $X/\langle W, u\rangle$ has conductor $DN$ and the Fourier coefficients of the $W$-fixed newform of level $DN$ with quaternionic $u$-eigenvalue $+1$, at every good prime below a hundred.
(c) The $W$-fixed newforms of level $DN$ split $1 + 2$ under $u$.
(d) The numerator of the zeta function of the model over $\mathbb{F}_p$, for good $p \le 60$, equals the one assembled from the trace formula's counts over $\mathbb{F}_p, \mathbb{F}_{p^2}, \mathbb{F}_{p^3}$, which determine it; likewise for the quotient.

**Status.** The construction is numerical and the checks are arithmetic consequences of the isogeny class and of the Galois structure of the branch locus; none of them proves the model. The strongest evidence available is agreement with a model obtained exactly. At $X_0^{1326}(1)/\langle w_{17}, w_{78}\rangle$, whose Borcherds level lies beyond the reach of the project's lattice-point enumerator, the double cover of the elliptic curve $X/\langle W, w_2\rangle$ branched at the CM points prescribed by the heights of Heegner points gives the exact model

$$y^2 = 272x^8 - 72x^6 - 55x^4 - 18x^2 + 17,$$

and the model recognised from the values of the three differentials at fourteen sample points, with $u = w_2$, is isomorphic to it over $\mathbb{Q}$. That is one instance; it is the one in which a numerical model produced here has met an exact one.

## 7. Two traps

**Precision must rise with the cutoff.** Raising the cutoff stops helping once the truncation error falls below the rounding error of the sum. At $X_0^{1326}(1)$ with thirty digits the residual of Noether's count was the same at cutoffs 46 and 50 while the rest of the spectrum agreed to twelve digits, which is exactly what a genuine small singular value would look like; raising the precision to fifty digits at cutoff 46 lowered it by a factor of a hundred, and the curve is hyperelliptic, as the exact model confirms. The sum there has some sixty million terms; above roughly ten million terms the working precision must rise with the cutoff. Two other explanations were tested first and ruled out: the fixed scaling by which the Gram matrix is rounded before the enumeration changes no digit (the scaling only decides which vectors are offered, and each is then accepted or rejected at full precision), and widening the margin on the Fourier coefficients changes no digit either.

**The Eichler order is not deterministic across processes.** Magma's construction of an Eichler order of level $N$ returns a different (conjugate) order from one process to the next, with the algebra's standard form and the real splitting unchanged. Values computed against different orders are values at different points of the curve, with different normalising constants, so a computation split over processes gives chunks that each lie on a canonical conic and share none. The order is therefore fixed once, through its $\mathbb{Z}$-basis in the standard basis of the algebra, and rebuilt in every process; with the same order the split computation reproduces the single-process invariants of the control to every digit. The same care is needed for the

Galois conjugates of a newform orbit: the generator of the coefficient field, and with it the order in which the conjugates are listed, is not the same from one process to the next either, so the conjugates are ordered by their coefficients.

## 8. What is implemented

The package `NelsonShimizu.m` provides: the Eichler order serialised and rebuilt; the basis of $W$-fixed differentials, newforms of level exactly $DN$ together with the oldform eigenvectors $v_\chi$, with the dimension check; the evaluation of the basis at a pair of points by (1) and (2), with the cutoff and precision as parameters; the direct measurement of the $W$-eigenvalues of every column; Noether's count with the reading rule above; the values on circles, their storage, and the model step through the elliptic quotient; and the four checks. Every docstring says that its result is numerical.

## References

[1] P. D. Nelson, *Evaluating modular forms on Shimura curves*, Math. Comp. **84** (2015), 2471–2503.

[2] J. Voight and J. Willis, *Computing power series expansions of modular forms*, in: Computations with modular forms, Contrib. Math. Comput. Sci. 6, Springer (2014), 331–361.